

Ahmed GHALEB

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 Chapter 3 Methodology

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CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

Development of the Al-Muneer Online Portal, with a defined set of requirements and scope identified, must be a systematic and structured process to ensure all objectives are fulfilled efficiently and effectively. From the very first concept and moving through the plan and design stages relevant to this PSM1 phase, the methodology taken must lead the project through all the way to its entire lifecycle before moving on to development and deployment within PSM2. The system development methodology taken for the Al-Muneer Online Portal, its stages, and how the methodology will be executed to deliver the goals of the project are described within this chapter. For a project of this type, with requirements well understood at the outset and where a structured approach is preferred, **the Waterfall model has been** recognized **as the** most appropriate guiding **methodology**. It offers a linear and structured methodology that is most relevant.



3.2 Methodology Choice and Justification

Because of the sequential and linear nature of software development along the Waterfall model, the model was selected for the Al-Muneer Online Portal project development. The model dictates segmentation of the project life cycle into phases, which should be finished before the commencement of the next phase. After initial coordination with stakeholders and proposal clarification, the Al-Muneer Online Portal project is in an ideal position to harness the advantages of such a stringent methodology, which is best suited for projects whose requirements are clearly defined and documented in the initial stage of the project. The clear-cut scope for functionalities like the interactive calendar, display of prices, upload of payment

proof, feedback system, easy reporting, and notification makes the Waterfall model a fitting selection.

selection.

Waterfall-style life cycles would normally be preferred in those projects where predictability and documentation detail matter most. The visibility of the phases of the Waterfall model provides established markers for measuring progress and outputs, i.e., System Design Document (SDD) and Software Requirements Specification (SRS), in the context of an FYP student project with distinct academic milestones (Progress 1, Progress 2, Final Report). Although Agile methodologies provide adaptability for projects with changing requirements, the Al-Muneer Online Portal is more advantageous under the Waterfall model's focus on preliminary planning and design, which reduces uncertainty as the project moves from the planning and design phases in PSM1 to development in PSM2. This strategy guarantees that the development work in PSM2 can advance based on a stable and mutually agreed foundation, which is vital for controlling the project scope within the academic timeline. The sequential structure also facilitates project management for an individual developer.

3.2.1 Waterfall Model Process Diagram

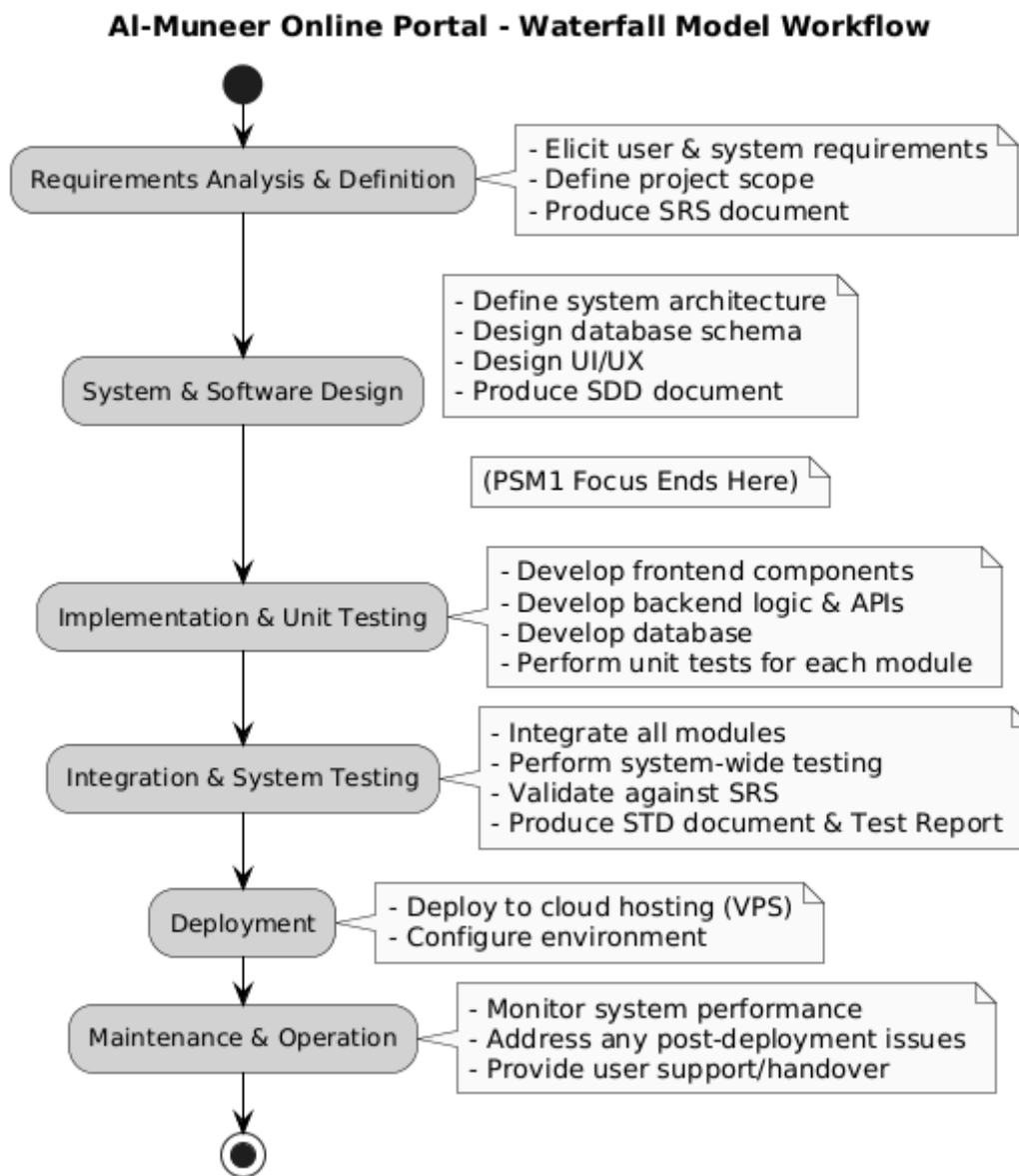


Figure 3.1: Waterfall Model Workflow for the Al-Muneer Online Portal

3.3 Phases of the Chosen Methodology

The development lifecycle for the Al-Muneer Online Portal, following the Waterfall model, is composed of distinct phases. Each phase has specific objectives and deliverables, ensuring a systematic progression towards the final product.

1. Requirements Analysis and Definition

This initial phase is essential for comprehending and documenting the specific requirements of the Al-Muneer Online Portal. It entails comprehensive discussions with the stakeholder (Mr. Almunajid) to extract all functional and non-functional requirements. Activities encompass identifying user roles (Visitor, Admin), defining features such as the display of venue information, media gallery, interactive availability calendar, pricing customization panel, submission of booking inquiries, simple payment proof uploads, feedback submissions, admin management of content, updates on payment status, review of feedback, basic report generation, and configuration of notifications. The primary output of this phase is the Software Requirements Specification (SRS) document, which acts as a foundational agreement regarding the functionalities of the system. This phase constituted a significant aspect of the early PSM1 activities.

2. System and Software Design

Following the clear definition and documentation of requirements in the SRS, the design phase commences. During this phase, the overall system architecture for the Al-Muneer Online Portal is formulated. This includes the design of the database schema (for MySQL/PostgreSQL), detailing the backend structure (Spring Boot services and APIs), and crafting the user interface (UI) and user experience (UX) for both the visitor-facing front-end and the administrator panel. Decisions regarding module breakdown, data flow, and security considerations are made at this stage. The principal deliverable for this phase is the System Design Document (SDD), which serves as a blueprint for the development team (the student developer). This

phase is a fundamental element of PSM1, particularly in preparation for Progress 2.

3. **Implementation and Unit Testing**

This stage signifies the initiation of actual coding and development, which will predominantly take place during PSM2. In accordance with the SDD, the Al-Muneer Online Portal will be constructed. The web application will be developed as a cohesive, singular project. The backend logic will be managed by Spring Boot (Java), which will also provide the user interface. The View layer will be created utilizing a server-side templating engine (such as Thymeleaf) along with HTML, CSS, and JavaScript. The database will be established and filled with initial structures. As each module or component is crafted, it will undergo unit testing to confirm that it operates correctly in isolation, adhering to its design Description.

4. **Integration and System Testing**

Once individual units have been developed and tested, they are combined into a comprehensive system. This integrated system subsequently undergoes thorough system testing to verify that all components function together seamlessly and that the system as a whole fulfills the requirements outlined in the SRS. This encompasses functional testing, usability testing, basic performance testing, and security testing. A Software Test Document (STD) will direct this process, and a test report will encapsulate the findings. This phase is also scheduled for PSM2.

5. **Deployment**

Al-Muneer Online Portal will be migrated into a production-like environment after system testing is successfully done and has been certified to be stable and in compliance with all requirements. Deployment onto a Cloud VPS is expected to be for this project. This stage includes putting the hosting

environment in place, deploying application code and database, and performing final testing to ensure the portal is live and user-accessible. This is expected to be finished towards the end of PSM2.

6. Maintenance and Operation

Subsequent to deployment, the maintenance phase commences. This phase entails the continuous monitoring of the Al-Muneer Online Portal for any potential issues, bugs, or necessary updates. Within the context of the FYP, this phase may involve a period of observation, rectifying any immediate post-deployment issues, and preparing documentation for the transition of the system to the stakeholder. Although extensive long-term maintenance falls outside the scope of an FYP, it is essential to ensure that the system is stable and functional upon delivery.

3.3.1 Gantt Chart for PSM1

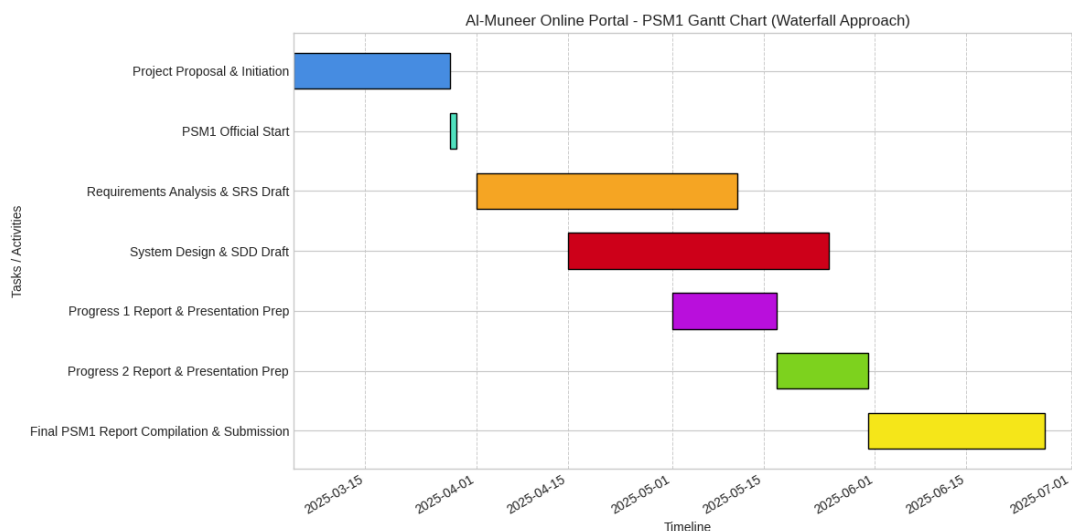


Figure 3.2: Gantt Chart for PSM1 Activities

3.3.2 Gantt Chart for PSM2

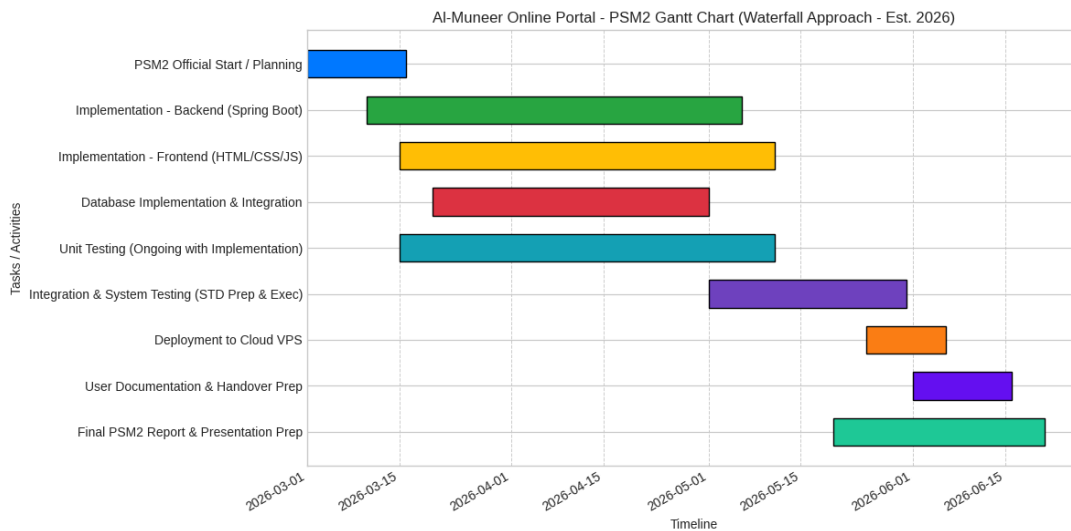


Figure 3.3: Proposed Gantt Chart for PSM2 Activities

3.4 System Requirements Analysis: Hardware and Software

The effective creation and implementation of the Al-Muneer Online Portal will rely on particular hardware and software resources. For the development environment, the essential hardware requirement is a relatively modern personal computer or laptop. This device should have adequate processing power (for instance, an Intel Core i5 or AMD Ryzen 5 equivalent, or superior), a minimum of 8GB of RAM (with 16GB recommended for enhanced multitasking with development tools), and sufficient storage capacity (such as a 256GB SSD or larger) to support the operating system, development tools, project files, and local database instances. The software prerequisites for development encompass a compatible operating system (Windows, macOS, or Linux), a Java Development Kit (JDK) version that aligns with Spring Boot (for example, JDK 11, 17, or 21), an Integrated Development Environment (IDE) like IntelliJ IDEA, Eclipse, or VS Code equipped with Java extensions, a build tool such as Maven or Gradle (usually managed by Spring Initializr), a relational database management system (MySQL Community Server or PostgreSQL installed locally for development and testing), a web browser

for testing purposes (for example, Chrome or Firefox), and version control software (Git).

For the deployment environment, a Cloud Virtual Private Server (VPS) will be utilized as the hardware. This VPS must possess specifications that include a minimum of 1-2 vCPUs, 2GB-4GB of RAM, and adequate SSD storage (for instance, 20GB-50GB) to accommodate the application, database, and any uploaded files such as payment proofs. The operating system of the server will generally be a Linux distribution (for example, Ubuntu or CentOS). The software stack on the deployment server will consist of the Java Runtime Environment (JRE) that is compatible with the developed application, the selected relational database server (either MySQL or PostgreSQL), a web server or application server if an embedded server from Spring Boot is not being utilized (although Spring Boot typically employs an embedded Tomcat, Jetty, or Undertow), along with any essential utilities for server management, monitoring, and security (such as firewall and SSH). Furthermore, an SSL/TLS certificate will be necessary to enable HTTPS.

3.5 Technology Used Description

The creation of the Al-Muneer Online Portal, utilizing the Waterfall methodology, systematically incorporates several essential technologies throughout its various phases. In the System and Software Design phase (PSM1), the architecture delineates the interactions among these technologies. The backend, developed with Spring Boot (Java) following an MVC pattern, will contain the fundamental business logic. This encompasses modules for overseeing venue information, managing booking inquiries, processing payment confirmations, handling feedback, generating reports, and initiating notifications. Spring Security will play a crucial role in safeguarding the admin panel.

The MySQL/PostgreSQL relational database, which is designed during this phase, will retain all persistent data, including hashed user credentials, hall specifications, availability, pricing structures, inquiry logs, payment proof details, and feedback entries. Its organized structure will directly facilitate the data retrieval necessary for the reporting functionalities required by the admin.

The user interface is set to be developed as the View component within the MVC architecture. It will utilize HTML5, CSS3, and JavaScript, with rendering occurring on the server-side through a templating engine (for instance, Thymeleaf) that is managed by Spring Boot. This methodology fosters a closely integrated application where the backend has direct control over the UI, which aligns perfectly with the objectives of this project.

Notification mechanisms will be crafted with a focus on WhatsApp channels, utilizing admin-triggered pre-filled messages to ensure a seamless flow of communication between the administrator and clients. This approach leverages WhatsApp deep-linking to facilitate quick manual alerts for booking confirmations and updates, aligning with local communication preferences while maintaining cost-efficiency. Additionally, email libraries available within Spring Boot may facilitate minimal email confirmations if deemed necessary.

The deployment phase will leverage a Cloud VPS, where the packaged Spring Boot application alongside the database will be hosted. Secure communication will be mandated via HTTPS (SSL/TLS) to safeguard all data transmitted between client browsers and the server. The Waterfall model guarantees that each of these technological components is designed, implemented, and subsequently tested in a systematic and orderly fashion.

3.6 Chapter Summary

This chapter gives a comprehensive overview of the System Development Methodology used in the Al-Muneer Online Portal, with special focus on the Waterfall model as its bedrock. The reason for using this methodology is that it is a structured, step-by-step approach that is better suited when the project has clearly defined requirements and educational deliverables. We went through the different phases of the Waterfall model as used in this project, beginning with Requirements Analysis and Definition through Deployment and Maintenance, with specific tasking assigned to PSM1 and PSM2. Visual Gantt charts provided timeframes for these phases. The chapter also outlined the software and hardware requirements for the development and deployment environments of the portal. It further detailed how the technologies used, like Spring Boot, a relational database, and traditional web frontend technologies, would be used and combined perfectly in the development process per the Waterfall methodology, so that there was a sequential and systematic approach towards the achievement of the project objectives.